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Internship Assessment Task: Python Development

Role: Software Engineering Intern

Last Submission Date: 10th May 2026

Objective: Most systems store and transmit data in bytes, and one byte can only hold values from 0 to 255. So a sensor value like 2850 simply doesn't fit in a single byte, it needs to be split into two bytes and reassembled on the other end.

You'll find this pattern in Bluetooth devices, Arduino sensors, and serial communication. Your task is to implement this split-and-reconstruct logic in Python and build a small real-time monitor around it. No hardware knowledge needed.

The Task

Part 1 - Split and Reconstruct

Write `encode(value)`

- Input: an integer from 0 to 4095
- Split using bitwise operators only:
 - `high = value >> 8`
 - `low = value & 0xFF`
- Return (high, low)

Write `decode(high, low)`

- Input: two integers (each 0-255)
- Reconstruct: `value = (high << 8) | low`
- Return the original integer

Test it:

Loop through 10 random values (0-4095), `encode` → `decode` → assert they match.
Print each result:

Original: 2850 → High: 11, Low: 34 → Decoded: 2850 ✓

Part 2 - Real-Time Monitor (2 Threads)

Build a simulation with two threads running at the same time.

Thread 1 - Sensor:

- Every 100ms, generate a random value (0-4095)
- Encode it and put (high, low) into a queue.Queue

Thread 2 - Monitor:

- Read (high, low) from the queue and decode it
- Maintain a running average of the last 10 values using `collections.deque(maxlen=10)` (it automatically drops the oldest value when full)
- If the running average goes above 3000 or below 500, print a warning:

WARNING: Running avg = 3142.0 - too high!

WARNING: Running avg = 423.5 - too low!

After 20 seconds, stop both threads cleanly and print:

Total values received: 198

Overall average: 2041.3

Threshold warnings: 14

Guidelines & Constraints

1. Documentation: Upside Down Labs official documentation (<https://docs.upsidedownlabs.tech/>) to know more about our products, its technical details, that will be asked during the interview process.
2. AI Usage: You may use AI to understand the concepts and workflow, but do not use it for code generation, those applications will be rejected if found.

Deliverables & Submission

Please organize your files clearly with version control. Submit your work as a GitHub repository, as we heavily utilize Git and Markdown for our open-source documentation and readme.

1. GitHub Repository Name – For uploading the files:

- Create a master/main repository with title:
UDL - [First Name] - [Roll Number] - Python

2. Your GitHub repository must include:

- 2.1 Encode_decode.py:- `encode()` and `decode()` functions + the 10-value test loop (Part 1)
- 2.2 monitor.py:- the two-thread simulation (Part 2)

3 analysis.md:- Written answers to all 5 questions (mandatory)

Ready to Submit?

Complete your application and drop your final task link into our official Google Form:

<https://forms.gle/4N9QhVx6JD7zjH9L9>

analysis.md - 5 Questions

Answer in your own words. Short and honest is better than long and copied.

1. Why can't 2850 fit in one byte? What is the maximum value a single byte can hold, and why?
2. For value = 2850, show step-by-step what "value >> 8" and "value & 0xFF" give you.
3. Explain what "(high << 8) | low" do during decoding. How do they reverse what encode() did?
4. What is the difference between a running average of the last 10 values and the total average of all values so far? Why does it matter for live data?
5. Why use queue.Queue between threads instead of a plain Python list?

Next Steps: The Interview

Your submission will be one of the core focuses of the technical interview. We look at correct encode/decode functions, proper bitwise ops, clean threading, correct running average logic, and honest written answers. Good luck!

Terms & Conditions

- Original Work: By submitting this task, you confirm that the code and documentations are solely your work.
- Evaluation: This assessment task designed solely to evaluate your technical fit for the internship role.
- Support: For any queries or doubts, you may mail us back by using the "reply all" option.

Good luck for the task & Keep tinkering.